

Homework 4

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Problem 1. Consider $f : A \rightarrow B$ a function from the set A to the set B , E subset of B , and F and G subsets of A .

1. Give the mathematical definition of the direct image of F under f .
2. Give the mathematical definition of the inverse image of E under f .
3. Show that

$$f(E \cup G) = f(E) \cup f(G)$$

Solution. 1. $f(F) = \{f(x) : \forall x \in F\}$

2. $f^{-1}(E) = \{x \in A : f(x) \in E\}$

■

Claim 1c. Show that

$$f(E \cup G) = f(E) \cup f(G)$$

Proof. $\forall b \in f(E \cup G)$, by definition of image, $\exists a \in E \cup G \ni f(a) = b$. Thus by definition of union, $a \in E$ or $a \in G$. If $a \in E$, then $b = f(a) \in f(E)$, by definition of image. If $a \in G$, then $b = f(a) \in f(G)$, by definition of image. Thus $b \in f(E)$ or $b \in f(G)$. Thus by definition of union, $b \in f(E) \cup f(G)$. Thus $f(E \cup G) \subseteq f(E) \cup f(G)$

$\forall b \in f(E) \cup f(G)$, by definition of union, $b \in f(E)$ or $b \in f(G)$. If $b \in f(E)$, by definition of image, $\exists a \in E \ni f(a) = b$. If $b \in f(G)$, by definition $\exists a \in G \ni f(a) = b$. Thus $a \in E$ or $a \in G$. Thus $a \in E \cup G$, thus by definition of image, $b \in f(E \cup G)$. Thus $f(E) \cup f(G) \subseteq f(E \cup G)$. Thus $f(E) \cup f(G) = f(E \cup G)$. □

Problem 2a. Give the definition for a set S to be denumerable

Proof. S is denumerable if $\exists f : \mathbb{N} \rightarrow S$ where f is a bijection. □

Claim 2b. Using only the definition, show that $\mathbb{Z}$ is denumerable.

Proof. $f : \mathbb{N} \rightarrow \mathbb{Z}$ where $\forall n \in \mathbb{N}$,

$$f(n) = \begin{cases} -\lfloor \frac{n}{2} \rfloor & \text{if } n \text{ is odd} \\ \frac{n}{2} & \text{if } n \text{ is even} \end{cases}$$

If $y > 0$, then $\exists n \in \mathbb{N} \ni n = 2y$, thus $f(n)\frac{2y}{2} = y$. Thus f is surjective.

$\lfloor \frac{a}{2} \rfloor = \lfloor \frac{b}{2} \rfloor \implies \frac{a}{2} = \frac{b}{2} \implies a = b$. If a, b are even, then $\frac{a}{2} = f(a) = f(b) = \frac{b}{2}$, thus $a = b$. Thus f is injective.

Claim 2c. *Show that $\mathbb{Q}$ is countable*

$\mathbb{Z} \times \mathbb{N}$ where you are spiraling around the origin, so you start with $(0, 0), (1, 0), (1, 1), (0, 1), (-1, 0), (-1, 0)$. Since we are going a finite number of elements before we change directions, we don't ever run out of $\mathbb{N}$. Moreover since we are spiraling while making sure to land on lattice points all values in $\mathbb{Z} \times \mathbb{N}$ will be covered. Thus $\exists$ a bijection $\mathbb{Z} \rightarrow \mathbb{N} \times \mathbb{N}$ and thus exists inverse bijection $g : \mathbb{N} \times \mathbb{Z} \rightarrow \mathbb{N}$.

$\forall r, s \in \mathbb{Q} \ni (g \circ f)(r) = (g \circ f)(s)$, since g is a bijection it is also injective so $f(r) = f(s)$, since f is injective, $r = s$. Thus $(g \circ f)$ is injective map from $\mathbb{Q} \rightarrow \mathbb{N}$. Thus $\mathbb{Q}$ is countable. $\square$

Problem 3. $\forall n \in \mathbb{N}$:

$$\sum_{i=1}^n \frac{1}{i(i+1)} = \frac{1}{1(2)} + \frac{1}{2(3)} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

Proof. (Base Case): Let $n = 1$, $\sum_{i=1}^1 \frac{1}{i(i+1)} = \frac{1}{1(2)} = \frac{1}{2}$. $n + 1 = 2$. So $\sum_{i=1}^1 \frac{1}{i(i+1)} = \frac{n}{n+1}$.

Inductive Hypothesis: Assume $\sum_{i=1}^n \frac{1}{i(i+1)} = \frac{n}{n+1}$ is true.

Inductive Step: $\sum_{i=1}^{n+1} \frac{1}{i(i+1)} = \sum_{i=1}^n \frac{1}{i(i+1)} + \frac{1}{(n+1)(n+2)}$. By Inductive Hypothesis,

$$\begin{aligned} \sum_{i=1}^n \frac{1}{i(i+1)} + \frac{1}{(n+1)(n+2)} &= \frac{n}{n+1} + \frac{1}{(n+1)(n+2)} \\ &= \frac{n(n+2)+1}{(n+1)(n+2)} \\ &= \frac{n^2+2n+1}{(n+1)(n+2)} \\ &= \frac{(n+1)^2}{(n+1)(n+2)} \\ &= \frac{n+1}{n+2}. \end{aligned}$$

Thus, $\sum_{i=1}^{n+1} \frac{1}{i(i+1)} = \frac{n+1}{n+2}$. Let $k = n + 1$. Thus, $\sum_{i=1}^k \frac{1}{i(i+1)} = \frac{k}{k+1}$.

Thus by induction, $\forall n \in \mathbb{N}$, $\sum_{i=1}^n \frac{1}{i(i+1)} = \frac{n}{n+1}$

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Problem 4. Using only the nine fundamental algebraic properties of $\mathbb{R}$, show that:

1. For all $a \in \mathbb{R}$, $-(-a) = a$
2. For all $a, b \in \mathbb{R}$, $(-a)(-b) = ab$.

Claim 4a. For all $a \in \mathbb{R}$, $-(-a) = a$

Proof. $\forall a \in \mathbb{R}$, by the existence of a additive inverse for all reals, $\exists -a \in \mathbb{R}$, $a + (-a) = 0$. By the commutative property of addition, $(-a) + a = 0$. Again by the existence of additive inverses $\exists -(-a) \in \mathbb{R}$ $\ni -(-a) + (-a) = 0$. Thus $0 + a = -(-a) + (-a) + a = -(-a)$. By properties of the additive identity, $a = 0 + a = -(-a)$. Thus $a = -(-a)$. $\square$

Claim 4b. For all $a, b \in \mathbb{R}$, $(-a)(-b) = ab$.

Proof. $\forall r \in \mathbb{R}$, by the properties of the multiplicative identity, $r * 1 = r$. By the properties of additive identity, $1 + 0 = 1$. Thus $r(1 + 0) = r$. By the distributivity of multiplication over addition, $r * 1 + r * 0 = r$. By the properties of the additive identity, $r + r * 0 = r$. By the existence of a additive inverse, $\exists -r \in \mathbb{R}$ $\ni -r + r = 0$. Thus $0 + r * 0 = -r + r + r * 0 = -r + r = 0$. By the properties of the additive identity, $0 + r * 0 = r * 0 = 0$. Thus $r * 0 = 0$.

$\forall a, b \in \mathbb{R}$. By the distributivity of multiplication over addition, $b(-1 + 1) = -1(b) + b$. $1 - 1 = 0$. Thus $b(0) = -1(b) + b$. By the above, $b(0) = 0$. Thus $-1(b) + b = 0$. By the existence of the additive identity, $-1(b) + b - b = -b \implies -1(b) = -b$.

Thus $(-a)(-b) = (a)(-1)(b)$. By the commutativity of multiplication, $(-a)(-1)(b) = (-1)(-a)b$. By the above, $(-1)(-a)(b) = -(-a)(b)$. By 4a, $-(-a) = a$. Thus $-(-a)(b) = ab$. Thus $(-a)(-b) = ab$. $\square$